Beyond Complexity: Do Resilience-Aware Factors Improve Consensus in Self-Evolving Distributed Networks?

Abstract

Purpose. Self-evolving communication networks can rewire links to keep distributed consensus working under failures. We ask whether a resilience-aware design, TCCE-R, which combines a consensus-error signal with graph-structural fragility (bridge fraction) and link-state volatility, accumulates them into a "resilience debt", and adapts its trigger using counterfactual feedback, improves consensus over simpler rewiring policies.

Methods. We evaluated eight policies in a simulator under one frozen, pre-specified protocol: three synthetic scenarios (random geometric graphs with 40 or 80 nodes; transient, background and spatially correlated burst failures) and two rewiring budgets (M = 3 and 6). Mutation thresholds were calibrated on separate seeds using mutation counts only, frozen with a SHA-256 manifest, and evaluated once on 60 held-out seeds per condition (2,640 of 2,700 planned runs; one arm in one condition could not be calibrated). Arms were compared paired by seed with Wilcoxon signed-rank tests, Holm correction, bootstrap confidence intervals, and a practical-effect margin of 0.02 in consensus error.

Results. Full TCCE-R (A5) was not shown to be better than the simple reactive threshold (A1): in four of the five evaluable conditions the two were practically equivalent, and in one (S2, M = 3) A1 had slightly lower error ($\Delta E = +0.0105$, below the margin). A state-blind front-loaded schedule (A7) had lower consensus error than A5 in all six conditions, by 0.024 to 0.076 in mean paired difference (Holm $p \leq 0.0031$; all point estimates exceed the 0.02 margin, though the CI lower bound is just below it in two conditions). Within the accumulated-debt family, adding connectivity and volatility signals without adaptation (A4 vs A3) did not help, and adaptive thresholds with counterfactual feedback (A5 vs A4) made no detectable difference. These results are specific to the simulated conditions and do not isolate the individual resilience signals, which were not ablated.

Conclusion. Under this protocol, resilience-aware control logic did not outperform simpler alternatives, and the timing of the first rewirings appears to matter more than the trigger's sophistication. We report this as a bounded negative result together with the protocol, code and diagnostics needed to audit it.

Keywords: distributed consensus; self-evolving networks; topology adaptation; network resilience; algebraic connectivity; negative results

1 Introduction

Distributed consensus lets network nodes agree through local exchanges and supports sensor fusion, coordination, and distributed estimation [1,2]. Its convergence depends on the communication graph, including its algebraic connectivity λ_2 [3,4]. Link failures change the effective graph and can degrade consensus [5]. Topology adaptation and edge rewiring have therefore been studied in adaptive coevolutionary

networks, topology control, virtual-topology adaptation, and connectivity-oriented graph design [6,7,8,9,10,11].

We test whether resilience-aware triggering improves on simpler topology-adaptation policies. TCCE-R (Topology–Consensus Co-Evolution with Resilience Debt) combines consensus error with structural fragility, link volatility, accumulated debt, and adaptive feedback. The evaluation does not establish consistent superiority over a reactive baseline; a state-blind front-loaded schedule achieved lower consensus error than TCCE-R in all six evaluated conditions.

Research questions.

- RQ1. Does full TCCE-R (A5) reduce consensus error relative to a reactive threshold policy (A1)?
- RQ2. Does front-loaded rewiring (A7) differ from adaptive TCCE-R (A5)?
- RQ3. Do the connectivity (C) and volatility (V) signals add value beyond a consensus-error-only accumulated debt (A3 → A4), and does adaptive thresholding with counterfactual feedback add value beyond that (A4 → A5)?
- RQ4. What trade-offs appear among consensus error, connectivity, recovery time and reconfiguration activity?

Contributions. (i) A reproducible protocol with separate calibration seeds, frozen configuration hashes, and one held-out evaluation; (ii) a matched-budget comparison of eight timing policies sharing the same candidate selector and safety gate; and (iii) a bounded negative result accompanied by exploratory diagnostics and a check of regression-to-the-mean bias in a legacy feedback rule. We make no novelty claim beyond this combination and do not claim that individual signals are ineffective.

2 Background and Related Work

Consensus and connectivity. In average consensus each node repeatedly replaces its value by a weighted combination of its neighbours' values; the weight matrix determines the asymptotic rate [2], and for fixed weights the rate is tied to the second-smallest Laplacian eigenvalue λ_2, the algebraic connectivity [3]. Switching topologies and delays have been analysed in [4], and nearest-neighbour coordination in [12]. Random link failures and noise were treated in [13,5], which also address topology design under random links. Robust average-consensus sensor fusion is considered in [14], and resilience to misbehaving nodes in [15]. Our setting differs: failures are benign (outages, bursts), nodes are honest, and the controller changes the topology.

Topology adaptation and rewiring. Edge rewiring to improve robustness or algebraic connectivity is studied in [9,10], and greedy or convex methods for growing well-connected graphs in [11]. Our selector uses a first-order estimate of the change in algebraic connectivity based on the Fiedler vector [11]. Connectivity control for mobile robot networks is surveyed in [16], and topology control for wireless ad hoc networks in [7]. Virtual-topology adaptation under dynamic traffic [8,17] motivates our interest in adapting topology as network conditions change.

Event- and change-triggered control. Triggering policies that act only when a signal warrants it are established in control [21,22]. Our accumulated debt is Page's one-sided CUSUM [23] applied to a stress signal.

Statistical methodology. Paired nonparametric tests [24], Holm's sequentially rejective correction [25], the bootstrap [26] and equivalence-style reasoning with a smallest effect size of interest [27] underlie the analysis. Regression to the mean, a pitfall for before/after evaluations triggered by extreme values, is reviewed in [28].

Position. We know of no prior controlled comparison of these specific timing policies under a common selector and budget. This is based on a limited, non-systematic literature review and is not a novelty claim.

3 System Model and Problem Formulation

Network and failures. A network has n nodes in the unit square (n = 40 or 80) and a set of candidate links between node pairs within radio range. A subset is configured (initially links within an initial radius). Time is divided into epochs k = 0,…,79, each consisting of T = 60 consensus rounds. In each epoch a configured link is live if it is not failed and not in setup. Three failure processes act: (a) transient outages, each configured link being unusable in each round independently with probability 0.05; (b) background on/off failures per epoch with fail probability p_fail and repair probability 0.1; (c) spatially correlated bursts: in S1 and S3, a burst starts with probability 0.1 per epoch, fails every potential link whose midpoint lies within radius 0.15 of a uniformly random centre for 2–6 epochs. S2 has no bursts and a higher background failure rate (Table 2). A new link becomes usable only after a setup delay of 2 epochs; a removed link disappears immediately.

Consensus error. At the start of every epoch each node draws fresh initial values x(0). Over T rounds nodes apply Metropolis-weight averaging on the live graph of that round: for live neighbours i, j, W_ij = 1/(1 + max(d_i, d_j)), where d_i is the live degree, and W_ii = 1 − Σ_j W_ij. With $\bar{x}$ the mean of x(0), the normalised error is

$$e(t)=(\| x(t)-\bar{x}1\|)/(\| x(0)-\bar{x}1\|), \quad E_k = 1-\min(1,\max(0,(\ln e_k(T))/(\ln\varepsilon))), \quad \varepsilon=0.01 .$$

E_k ∈ [0, 1]: 0 means the error fell to ε or below within the epoch, 1 means no progress. Lower E is better. The primary outcome of a run is the mean of E_k over its epochs.

Control problem. A logically centralised controller observes the graph and the history of E_k, and may mutate the topology at most once per epoch and at most M times per run (budget). Every mutation is a rewire: it removes one configured link and adds one candidate link (strict edge budget). The question is when to attempt a mutation, with candidate selection and safety checks held identical across arms.

Scope of the implemented model. The controller is logically centralised (global graph information); we do not study distributed control. Node load, traffic conditions, trust/uncertainty and adversarial attacks, which appear in the original architecture sketch, are not implemented. R (resilience) is operationalised provisionally as λ_2 of the unweighted live-graph Laplacian; the term "resilience cost" in the original diagram is undefined, so M below is a ranking, not a benefit/cost ratio.

4 TCCE-R Architecture and Design

Figure 1 summarises the nine stages: network-state monitoring, consensus monitoring, resilience-debt estimation, adaptive triggering, candidate evaluation, safety checking, mutation selection, network reconfiguration, and feedback.

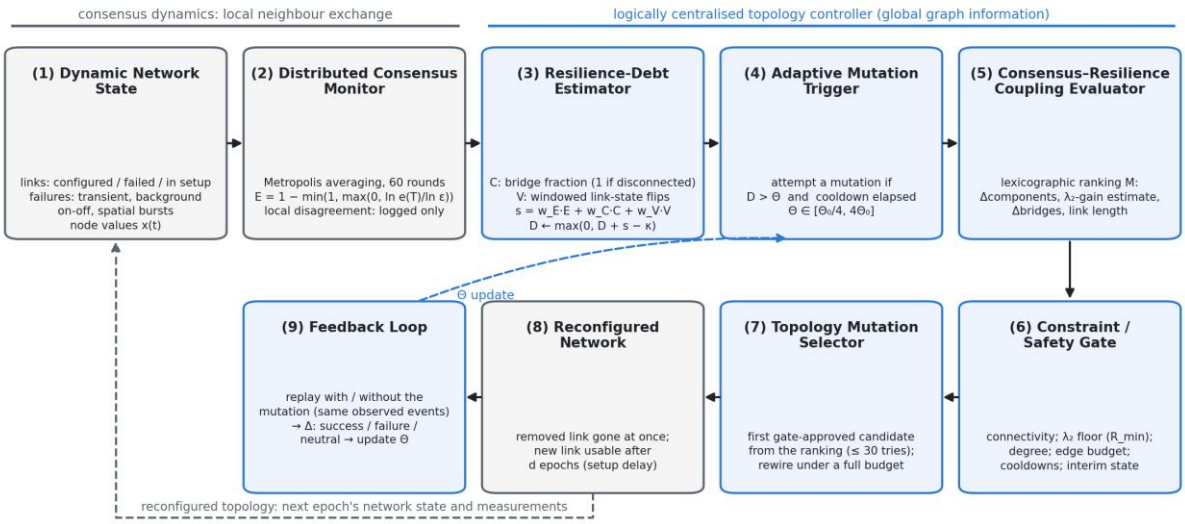


Figure 1. TCCE-R architecture as implemented. Blue stages implement the proposed mechanism or differ between compared strategies; grey stages are the shared environment. The controller is logically centralised. Not implemented as mechanisms: load, traffic, trust/uncertainty, attack model; R = λ_2 is provisional.

Signals. Let L be the set of live links of the epoch's graph and Q the set of configured links.

- Consensus error E_k (Section 3).
- Structural fragility: C_k = 1 if the live graph is disconnected, otherwise C_k = #bridges/|L|.
- Link volatility: V_k=min(1, (1)/(|Q|)Σ_(w=k-2)^kflips_w), where flips counts changes of up/down state of configured links over the last 3 epochs.

Stress and debt. The stress is s_k = w_E E_k + w_C C_k + w_V V_k. The resilience debt is Page's one-sided CUSUM,

$$D_k=\max(0, D_{(k-1)}+s_k-\kappa),$$

reset to 0 only after a successful mutation. In the evaluated arms, A3 uses s = E with reference κ_E; A4 and A5 use s = 0.5E + 0.3C + 0.2V with reference κ_s (weights fixed in code; no weight tuning was performed). κ_E and κ_s are the medians of E and of the stress over connected epochs of the static arm A0 on calibration seeds (Table 2).

Trigger. A mutation is attempted if D_k > Θ and at least 2 epochs have passed since the last successful mutation (the policy-level wait between mutations). In A5, Θ is adaptive within [Θ_0/4, 4Θ_0]: Θ ← 0.9Θ after a success, Θ ← 1.25Θ after a failure, and it is unchanged if the outcome is neutral.

Candidate ranking M. Candidate rewires are ordered lexicographically by (i) change in number of connected components, (ii) a first-order estimate of the gain in λ_2 for the added link, (f_i-f_j)^2 with f the Fiedler vector [11], (iii) change in bridge count, (iv) link length. The top 10 form a shortlist; gains within 10% of the best are treated as ties. This is a ranking, not a ratio.

Safety gate. A candidate is accepted only if all hold, in order: neither touched edge is within its 5-epoch edge-level cooldown; the added link is not currently failed; the edge budget is respected (every mutation is a pure addition-plus-removal); both endpoints keep degree ≤ 10 (configured basis); the number of components does not increase (checked on the final and interim setup state); and λ_2(after) ≥ min(R_min, λ_2(before)), with R_min = 0.05. The selector tries candidates in ranked order (at most 30) and applies the first accepted one.

Counterfactual feedback (A5). For a mutation at epoch m, the controller replays the window {m+1, …, m+d+H} (d = 2 setup epochs, H = 3 horizon) with and without the mutation under the identical observed failures, outages and initial values x(0). With Δ the mean of E(with) minus the mean of E(without), the outcome is success if Δ < 0, failure if Δ > 0, neutral if Δ = 0. Each mutation is evaluated in isolation and no future information is used. This replaces an earlier before/after rule that is biased by regression to the mean (Section 8.3). The replay assumes an exact model of the consensus dynamics.

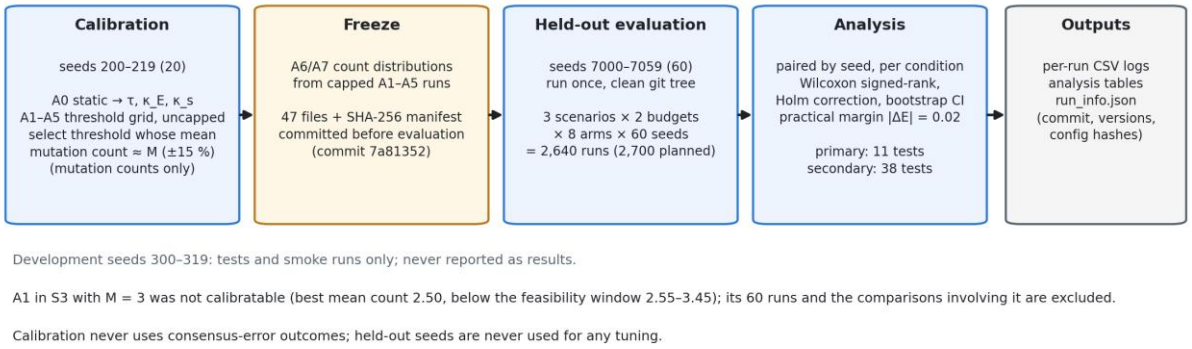


Figure 2. Study workflow: calibration on seeds 200–219 using mutation counts only, a freeze with a SHA-256 manifest, a single held-out evaluation on seeds 7000–7059, and a pre-specified paired analysis.

5 Experimental Methodology

Arms. Eight policies decide only when to attempt a mutation; they share the selector, gate, cooldown of 2, per-run cap M, one attempt per epoch, strict edge budget and setup delay (Table 1).

Arm	Name	Mutation attempted when	Stress / state used
A0	Static	never (reference; defines κ, τ, recovery anchors)	—
A1	Reactive	E_k > h	instantaneous E
A2	Moving average	4-epoch moving average of E > h	E
A3	CUSUM-E	D > Θ, with s = E and reference κ_E	E (accumulated)
A4	CUSUM-ECV	D > Θ, with s = 0.5E + 0.3C + 0.2V, reference κ_s	E, C, V (accumulated), fixed Θ
A5	Full TCCE-R	as A4, with adaptive Θ and counterfactual feedback	E, C, V, adaptive Θ

A6	Random timing	at random epochs (gaps ≥ 2), per-seed count from calibration distribution	none (state-blind)
A7	Front-loaded	at epochs 0, 2, 4, … until the per-seed count is reached	none (state-blind)

Table 1. Compared arms. All share the cooldown of 2 epochs, cap M, one attempt per epoch, the same selector, safety gate and strict edge budget.

Scenarios and budgets. Table 2 lists the three synthetic scenarios (S1 RGG-40 bursty, S2 RGG-40 background failures only, S3 RGG-80 bursty), each with 80 epochs × 60 rounds, at budgets $M \in \{3, 6\}$. A condition is a scenario–budget pair (six in total). They are synthetic; no real traces were used.

Scenario	Nodes	Initial link radius	Radio range (candidate links)	Transient loss / round	Background failure / repair (per epoch)	Bursts (rate; radius; duration)	Epochs × rounds	Setup delay (epochs)	τ (episode threshold)	κ_E	κ_s
S1	40	0.22	0.32	0.05	0.005 / 0.1	0.1/epoch; 0.15; 2–6 epochs	80 × 60	2	0.6352	0.4171	0.2297
S2	40	0.22	0.32	0.05	0.01 / 0.1	none	80 × 60	2	0.5803	0.4277	0.2369
S3	80	0.1556	0.2263	0.05	0.005 / 0.1	0.1/epoch; 0.15; 2–6 epochs	80 × 60	2	0.6485	0.4866	0.2640

Calibration. Thresholds (h for A1/A2, Θ for A3/A4, Θ_0 for A5) were chosen on calibration seeds 200–219 over a 40-point grid (h: 0.30–0.98 linear; Θ and Θ_0: 0.05–20 geometric) by picking, per condition, the value whose mean uncapped mutation count was closest to M. A threshold was feasible only within ±15% of M. No outcome metric (E, connectivity, recovery) was used. A6 and A7 draw their per-seed intended mutation counts from the empirical distribution of capped A1–A5 realised counts on calibration seeds. A1 in S3 with M = 3 was not calibratable (best mean count 2.50, below the window 2.55–3.45); its 60 runs were excluded, giving 2,640 of 2,700 planned runs, and every comparison involving it is omitted. All other arm–condition cells were feasible (Table A2). The calibration comprised 12,640 runs (the protocol planned 12,660; the 20-run difference equals the capped count-distribution runs of the one non-calibratable A1 cell). Configurations, constants and calibration report (47 files) were frozen with a SHA-256 manifest and committed (commit 7a81352, which is two commits after the repository tag v0.7 = f874ad7, the protocol commit) before evaluation.

Evaluation. Held-out seeds 7000–7059 (60 per condition) were run once from the frozen code snapshot identified by commit 7a81352 (Python 3.11.9, NumPy 2.3.4, SciPy 1.17.1, pandas 2.3.3; 272.3 seconds). Development seeds 300–319 were used only for tests and smoke runs.

Statistics. Within each condition arms are paired by seed (n = 60). For a comparison X − Y the difference is $\Delta E = E_X - E_Y$; negative favours X. We use two-sided Wilcoxon signed-rank tests (p = 1 if all differences are zero), Holm correction within the primary family (12 pre-specified comparisons, 11 evaluable) and the secondary family (42 pre-specified, 38 evaluable), 95% bootstrap CIs (10,000 resamples, seed 20261010), and rank-biserial effect sizes (in the supplementary analysis outputs). The practical-effect margin is $|\Delta E| = 0.02$. A verdict is meaningfully better (Holm $p < 0.05$ and $|\Delta E| \geq 0.02$), better below margin (Holm $p < 0.05$, $|\Delta E| < 0.02$), practically equivalent (Holm $p \geq 0.05$ and 95% bootstrap CI within ±0.02) or inconclusive. Conditions are never pooled. Primary comparisons: A5 − A1 and A5 − A7. Secondary: A2 − A1, A3 − A1, A3 − A2, A4 − A3, A5 − A4, A6 − A1, A7 − A1. Everything else (connectivity, λ_2, recovery, activity, mechanism, placebo diagnostics) is exploratory and carries no p-values.

Exploratory metrics. Disconnected-epoch fraction; mean λ_2; anchored recovery time: for each epoch s at which a disconnection episode of the static arm A0 (same seed) begins, the number of epochs t − s until the arm's own live graph is first connected at some epoch $t \geq s$ (right-censored at the end of the run, counted as 80 − s), averaged over that run's anchors (using the anchored-recovery calculation defined in the analysis protocol); runs without anchors are missing. The threshold τ in Table 2 is a 75th-percentile of A0's E used for episode labelling (tau_quantile = 0.75); realised mutations; and live-link releases (working links removed by rewiring).

6 Results

6.1 Primary comparisons (RQ1, RQ2)

Condition	Comparison (E_X − E_Y)	n	ΔE	95% bootstrap CI	Holm p	Reading
S1, M=3	A5 − A1	60	-0.0053	[-0.0131, +0.0019]	0.865	practically equivalent
S1, M=3	A5 − A7	60	+0.0448	[+0.0286, +0.0600]	1.27×10^{-5}	A7 lower E, ≥ margin
S1, M=6	A5 − A1	60	-0.0046	[-0.0103, +0.0012]	0.513	practically equivalent
S1, M=6	A5 − A7	60	+0.0756	[+0.0597, +0.0912]	5.45×10^{-9}	A7 lower E, ≥ margin
S2, M=3	A5 − A1	60	+0.0105	[+0.0027, +0.0182]	0.033	A1 lower E, < margin
S2, M=3	A5 − A7	60	+0.0461	[+0.0291, +0.0627]	4.04×10^{-5}	A7 lower E, ≥ margin
S2, M=6	A5 − A1	60	+0.0080	[+0.0008, +0.0151]	0.213	practically equivalent
S2, M=6	A5 − A7	60	+0.0381	[+0.0184, +0.0570]	0.003	A7 lower E, ≥ margin
S3, M=3	A5 − A7	60	+0.0244	[+0.0188, +0.0298]	4.76×10^{-8}	A7 lower E, ≥ margin

S3, M=6	A5 − A1	60	+0.0002	[-0.0035, +0.0036]	0.865	practically equivalent
S3, M=6	A5 − A7	60	+0.0321	[+0.0255, +0.0382]	1.04×10^{-8}	A7 lower E, ≥ margin

Table 3. Primary comparisons (ΔE = E_X − E_Y; negative favours X). Holm over the 11 evaluable of 12 primary tests. Figure 3 shows E by arm and Figure 4 the paired differences with the margin.

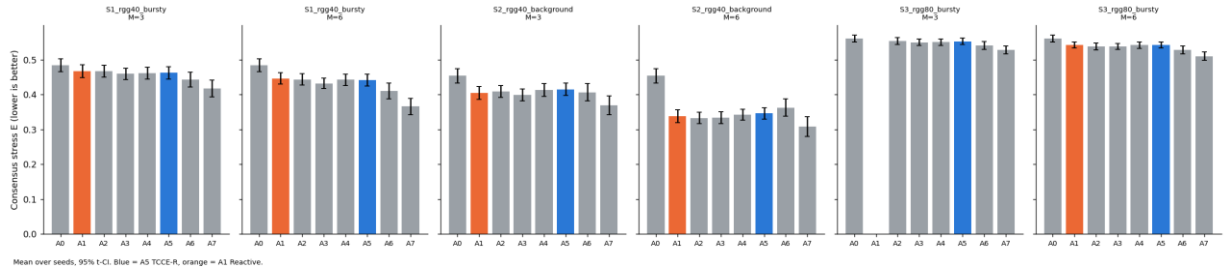


Figure 3. Mean consensus error E by arm and condition (frozen analysis output, unchanged). Lower is better.

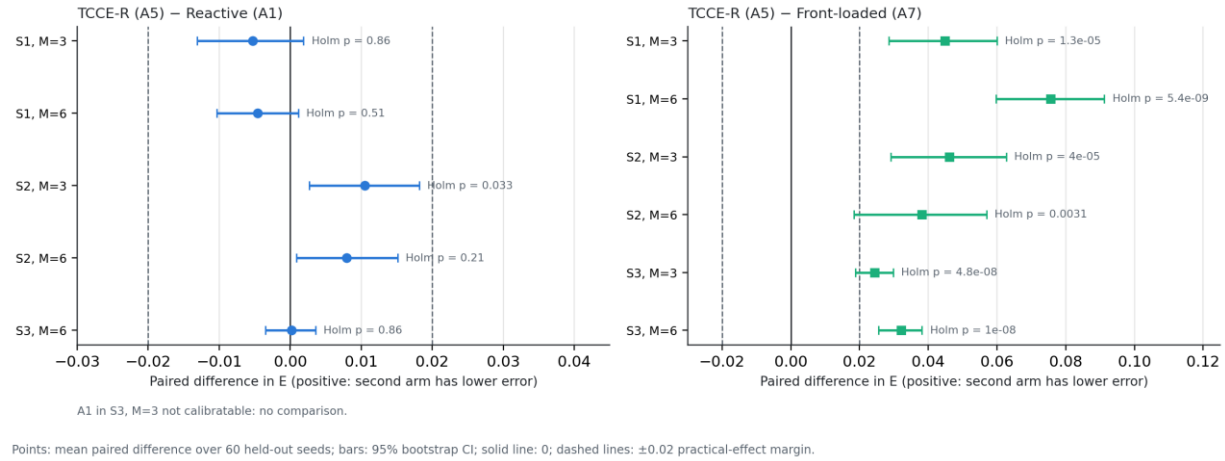


Figure 4. Primary paired differences with 95% bootstrap CIs. Dashed lines mark the ±0.02 practical margin. Positive values mean the second arm has lower error.

RQ1 (A5 vs A1). A5 − A1 was practically equivalent in four of five evaluable conditions (ΔE from −0.0053 to +0.0080); in S2 with M = 3, A1 had lower error by 0.0105 (Holm p = 0.033), below the practical margin. Note that for the practically equivalent S2, M = 6 case the bootstrap CI lower bound is +0.0008, close to zero but the Holm-adjusted test is not significant. We therefore find no evidence that A5 improves on A1, and no evidence that it is worse by a practically meaningful amount; the A1 S3 M = 3 condition could not be evaluated.

RQ2 (A5 vs A7). A7 had lower E than A5 in all six conditions (ΔE = +0.0244 to +0.0756), every mean difference exceeding the 0.02 margin with Holm p ≤ 0.0031. In S2 M = 6 and S3 M = 3 the 95% CI lower bounds (+0.0184 and +0.0188) fall slightly below the margin, so the direction of A7's advantage is clear in all six conditions but its size relative to the margin is less certain in those two. This does not mean A7 is universally superior: it is a state-blind schedule evaluated only in these simulated conditions, and it has costs (Section 6.4).

6.2 Secondary comparisons (RQ3)

Comparison (E_X − E_Y)	Evaluable conditions	Practically equivalent	X lower E, ≥ margin	X lower E, < margin (significant)	Y lower E, ≥ margin	Y lower E, < margin (significant)	Inconclusive	Range of ΔE	Smallest Holm p
A2 − A1	5	5	0	0	0	0	0	-0.0050 to +0.0040	0.690
A3 − A1	5	4	0	0	0	0	1	-0.0141 to -0.0038	0.058
A3 − A2	6	6	0	0	0	0	0	-0.0110 to +0.0006	0.060
A4 − A3	6	4	0	0	0	2	0	+0.0008 to +0.0144	0.003
A5 − A4	6	6	0	0	0	0	0	-0.0009 to +0.0034	0.307
A6 − A1	5	1	2	1	0	0	1	-0.0357 to +0.0245	1.42×10^{-4}
A7 − A1	5	0	5	0	0	0	0	-0.0802 to -0.0302	5.73×10^{-9}

Table 4. Secondary comparisons summarised by contrast over evaluable conditions (full per-condition results in Table A1).

- A3 vs A1 (accumulated debt on E alone vs instantaneous threshold): practically equivalent in four conditions, inconclusive in one.
- A4 vs A3 (adding C and V to the stress): A3 had lower E in two conditions (below margin, Holm p down to 0.003) and the contrast was equivalent in four; adding C and V did not help here.
- A5 vs A4 (adaptive Θ plus counterfactual feedback): practically equivalent in all six conditions (ΔE −0.0009 to +0.0034).
- A7 vs A1: A7 had lower E in all five evaluable conditions, all ≥ margin (ΔE −0.0802 to −0.0302).

- A6 vs A1: mixed (random timing sometimes better, sometimes not): two meaningful, one below-margin, one equivalent, one inconclusive.

These comparisons are not component ablations: C, V, and the adaptive-feedback mechanism were not removed individually from A5. Their individual effectiveness therefore remains undetermined.

6.3 Feedback quality

Condition	Updates per A5 run	Success	Failure	Neutral	Mean Δ (E_with − E_without)
S1, M=3	2.42	83.9%	15.5%	0.6%	-0.0234
S1, M=6	5.33	72.7%	26.1%	1.2%	-0.0158
S2, M=3	2.30	78.2%	20.3%	1.5%	-0.0410
S2, M=6	5.15	84.9%	14.3%	0.8%	-0.0347
S3, M=3	2.50	78.3%	21.1%	0.6%	-0.0097
S3, M=6	5.43	76.9%	22.4%	0.7%	-0.0064

Table 6. A5 counterfactual feedback outcomes (descriptive). Counterfactual replay judged 72.7%–84.9% of A5's mutations as successes, with mean Δ between −0.041 and −0.006, i.e. mutations reduced replayed E within their windows on average, yet A5's overall E was not better than A1's. These shares are not a performance claim: they measure a short-window local effect under an exact-model assumption and are not calibrated against a placebo.

6.4 Trade-offs (RQ4; exploratory)

Condition	Arm	E	Disconnected fraction	λ_2	Anchored recovery (epochs)	Realised mutations	Live-link releases
S1, M=3	A0	0.4834	0.362	0.050	6.17	0.00	0.00
S1, M=3	A1	0.4672	0.351	0.056	6.17	2.47	0.02
S1, M=3	A5	0.4619	0.334	0.058	5.44	2.50	0.17
S1, M=3	A7	0.4172	0.340	0.086	6.94	2.25	1.90
S1, M=6	A0	0.4834	0.362	0.050	6.17	0.00	0.00
S1, M=6	A1	0.4458	0.318	0.066	5.08	5.42	0.15
S1, M=6	A5	0.4412	0.306	0.069	4.66	5.42	0.32
S1, M=6	A7	0.3656	0.294	0.114	4.53	4.80	3.63
S2, M=3	A0	0.4536	0.289	0.054	10.40	0.00	0.00
S2, M=3	A1	0.4043	0.163	0.076	3.97	2.52	0.65
S2, M=3	A5	0.4148	0.169	0.069	3.86	2.33	0.40
S2, M=3	A7	0.3687	0.205	0.099	5.00	2.20	2.05
S2, M=6	A0	0.4536	0.289	0.054	10.40	0.00	0.00
S2, M=6	A1	0.3376	0.114	0.113	2.99	5.22	2.70
S2, M=6	A5	0.3456	0.113	0.107	2.61	5.20	2.62
S2, M=6	A7	0.3075	0.169	0.134	3.89	4.87	4.37
S3, M=3	A0	0.5606	0.456	0.017	7.34	0.00	0.00
S3, M=3	A1	—	—	—	—	—	—
S3, M=3	A5	0.5523	0.431	0.020	6.33	2.63	0.07
S3, M=3	A7	0.5279	0.425	0.031	6.13	2.37	1.80
S3, M=6	A0	0.5606	0.456	0.017	7.34	0.00	0.00
S3, M=6	A1	0.5419	0.419	0.023	5.98	5.50	0.17

S3, M=6	A5	0.5420	0.407	0.023	5.69	5.57	0.17
S3, M=6	A7	0.5100	0.413	0.039	6.03	5.12	3.47

Table 5a. Descriptive means over 60 seeds per condition (A0 Static, A1 Reactive, A5 TCCE-R, A7 Front-loaded; A1 absent in S3, M = 3).

Condition	Comparison	Δ disconnected	Δ λ_2	Δ recovery (epochs)	Δ live-link releases	ΔE, equal counts
S1, M=3	A5 − A1	-0.018 [-0.045, +0.005]	+0.002 [-0.001, +0.006]	-0.74 [-1.98, +0.13]	+0.15 [+0.05, +0.27]	-0.0047 [-0.0159, +0.0054] (n=39)
S1, M=3	A5 − A7	-0.006 [-0.046, +0.028]	-0.028 [-0.036, -0.019]	-1.50 [-4.10, +0.56]	-1.73 [-2.02, -1.43]	+0.0690 [+0.0433, +0.0919] (n=24)
S1, M=6	A5 − A1	-0.012 [-0.026, +0.003]	+0.003 [+0.000, +0.006]	-0.42 [-0.97, +0.04]	+0.17 [+0.05, +0.30]	-0.0022 [-0.0086, +0.0044] (n=42)
S1, M=6	A5 − A7	+0.011 [-0.008, +0.031]	-0.045 [-0.054, -0.035]	+0.12 [-0.48, +0.67]	-3.32 [-3.80, -2.82]	+0.1029 [+0.0803, +0.1257] (n=24)
S2, M=3	A5 − A1	+0.006 [-0.023, +0.032]	-0.007 [-0.011, -0.002]	-0.11 [-1.35, +0.85]	-0.25 [-0.47, -0.03]	+0.0044 [-0.0045, +0.0128] (n=42)
S2, M=3	A5 − A7	-0.037 [-0.079, +0.002]	-0.030 [-0.039, -0.020]	-1.14 [-2.40, +0.04]	-1.65 [-1.95, -1.32]	+0.0625 [+0.0413, +0.0828] (n=24)
S2, M=6	A5 − A1	-0.001 [-0.015, +0.012]	-0.006 [-0.010, -0.002]	-0.38 [-0.86, +0.02]	-0.08 [-0.33, +0.18]	+0.0019 [-0.0050, +0.0088] (n=48)
S2, M=6	A5 − A7	-0.057 [-0.094, -0.022]	-0.027 [-0.039, -0.015]	-1.27 [-2.33, -0.32]	-1.75 [-2.40, -1.03]	+0.0612 [+0.0437, +0.0788] (n=27)
S3, M=3	A5 − A7	+0.007 [-0.013, +0.028]	-0.011 [-0.014, -0.008]	+0.20 [-0.42, +0.83]	-1.73 [-2.03, -1.43]	+0.0311 [+0.0249, +0.0371] (n=29)
S3, M=6	A5 − A1	-0.011 [-0.028, +0.004]	+0.000 [-0.001, +0.002]	-0.29 [-0.82, +0.20]	+0.00 [-0.12, +0.12]	+0.0008 [-0.0043, +0.0050] (n=38)
S3, M=6	A5 − A7	-0.005 [-0.032, +0.018]	-0.016 [-0.019, -0.013]	-0.34 [-1.59, +0.61]	-3.30 [-3.72, -2.90]	+0.0370 [+0.0302,

						+0.0435] (n=28)

Table 5b. Exploratory paired differences (mean [95% bootstrap CI]; no p-values). The last column restricts to seeds with equal realised mutation counts in both arms (n shown).

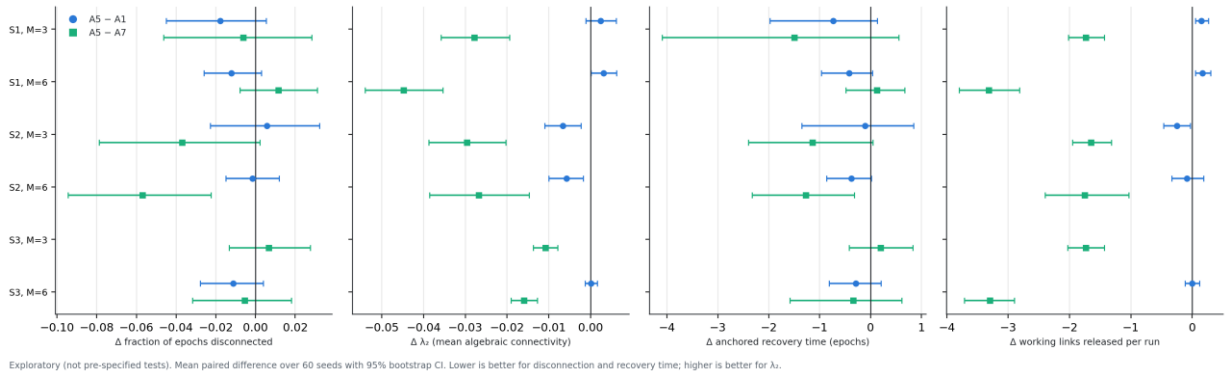


Figure 5. Exploratory paired differences in disconnection, λ_2, recovery and live-link releases.

- A5 vs A1. Differences in disconnected fraction, recovery time and mean λ_2 are small and their intervals mostly include zero. A5 has slightly lower λ_2 in S2 (−0.007, −0.006; intervals exclude zero) and releases slightly more working links in S1 (+0.15, +0.17).
- A5 vs A7. A7 achieves higher mean λ_2 in all six conditions (differences −0.011 to −0.045, intervals exclude zero), consistent with its lower E, at the price of releasing 1.65–3.32 more working links per run than A5 (all intervals exclude zero). In S2 (no bursts) A7 also had a larger disconnected fraction (A5 − A7 = −0.037 and −0.057, the latter interval excluding zero) and longer anchored recovery (−1.27 epochs for M = 6, interval excluding zero). So A7's lower E does not come with uniformly better connectivity or recovery.
- Equal realised counts. Restricting to seeds with equal counts gives similar conclusions (A5 − A7 ΔE +0.031 to +0.103; A5 − A1 within ±0.005).

Mechanism (exploratory). Table A3 shows that reactive and CUSUM-type arms mutate overwhelmingly while the network is disconnected (A1: 45–99%; A5: 47–98%), whereas A7 does so for 7–30%, and its first mutation is at epoch 0. A reconnecting move repairs a failure that may heal anyway and may be disproportionately wasted in bursts; an early move improves the baseline connectivity (λ_2) for the whole run. We offer this as a hypothesis consistent with the data, not as a tested causal result.

6.5 Supplementary replication (exploratory)

An earlier exploratory experiment using A1, A3 and A5, with separate calibration (seeds 400–419) and evaluation (seeds 500–559), is summarised in Table S1. A5 was practically equivalent to A1 in all evaluable cells; A3 had lower E than A5 in four cells, three below the practical-effect margin. This experiment was not pre-specified, used an earlier code/configuration snapshot, and did not include A6 or A7; it is therefore only a consistency check and cannot address the front-loading question.

Condition	Comparison (E_X − E_Y)	ΔE	95% bootstrap CI	Holm p	Reading
S1, M=3	A3 − A1	excluded	not calibratable	n/a	excluded

S1, M=3	A5 − A1	excluded	not calibratable	n/a	excluded
S1, M=3	A5 − A3	+0.0066	[-0.0007, +0.0144]	0.308	practically equivalent
S1, M=6	A3 − A1	-0.0229	[-0.0310, -0.0150]	6.89×10^{-6}	A3 lower E, ≥ margin
S1, M=6	A5 − A1	-0.0102	[-0.0195, -0.0011]	0.166	practically equivalent
S1, M=6	A5 − A3	+0.0127	[+0.0059, +0.0200]	0.009	A3 lower E, < margin
S2, M=3	A3 − A1	-0.0154	[-0.0268, -0.0051]	0.166	inconclusive
S2, M=3	A5 − A1	+0.0075	[-0.0011, +0.0164]	0.425	practically equivalent
S2, M=3	A5 − A3	+0.0228	[+0.0132, +0.0336]	1.63×10^{-4}	A3 lower E, ≥ margin
S2, M=6	A3 − A1	-0.0018	[-0.0098, +0.0069]	1.000	practically equivalent
S2, M=6	A5 − A1	excluded	not calibratable	n/a	excluded
S2, M=6	A5 − A3	excluded	not calibratable	n/a	excluded
S3, M=3	A3 − A1	-0.0051	[-0.0090, -0.0013]	0.166	practically equivalent
S3, M=3	A5 − A1	-0.0002	[-0.0043, +0.0038]	1.000	practically equivalent
S3, M=3	A5 − A3	+0.0049	[+0.0021, +0.0077]	0.002	A3 lower E, < margin
S3, M=6	A3 − A1	-0.0089	[-0.0129, -0.0049]	2.32×10^{-4}	A3 lower E, < margin
S3, M=6	A5 − A1	-0.0037	[-0.0086, +0.0012]	0.425	practically equivalent
S3, M=6	A5 − A3	+0.0052	[+0.0017, +0.0087]	0.008	A3 lower E, < margin

7 Discussion

What the evidence supports. Under this protocol, TCCE-R was not shown to outperform the reactive threshold, while front-loaded timing achieved lower consensus error in all six evaluated conditions. This is consistent with early reconfiguration mattering more than trigger complexity in these scenarios, but the experiment does not establish that explanation causally.

What it does not support. Because C, V and D were not ablated individually, the results do not show that any one signal is ineffective; A4 versus A3 tests adding C and V together with fixed weights and threshold. Nor do the results establish that A7 is generally superior: it is state-blind and incurs costs in live-link releases and, in S2, disconnection and recovery. Findings are limited to three synthetic scenarios, two budgets and one controller design; no pooled winner is declared.

Why might resilience-aware triggering not help? Three untested hypotheses are that debt responds after degradation begins, mutations attempted during disconnection may be wasted when transient failures heal, and short-window counterfactual feedback may not optimise long-run error. The mechanism analysis is consistent with the first two explanations but does not test causality.

Statistical vs practical significance. Several A5 − A7 differences are statistically significant and large relative to the practical-effect margin; the one significant A5 − A1 result is smaller than that margin. We therefore interpret confidence intervals alongside p-values.

8 Threats to Validity

1. Construct validity. E is a consensus-error proxy normalised per epoch with fresh initial values; R = λ_2 is provisional; M is a lexicographic ranking rather than a benefit/cost measure.
2. Internal validity. Arms are matched on mean, not per-seed, mutation counts (realised counts reported; equal-count analysis exploratory). A5 differs from A1 in several ways at once (debt accumulation, C/V terms, adaptive Θ, feedback), so differences cannot be attributed to a single factor. A5's weights are hard-coded in the policy (no ablation). The counterfactual replay assumes an exact model of the dynamics.
3. Regression to the mean. A legacy before/after feedback rule labelled 67–73% of no-op placebo mutations (Static arm) as "successes" versus 77–81% for A5's own updates (Table A4, v0.6 conditions), demonstrating a bias that motivated the counterfactual rule. The placebo comparison was computed on the earlier v0.6 data, not on v0.7.
4. External validity. Synthetic random geometric graphs, 40/80 nodes, particular failure rates, honest nodes, logically centralised control, one gate configuration.
5. Statistical. 60 seeds per cell give limited power for effects below about 0.02; Holm correction applied to evaluable tests only (11 of 12 primary, 38 of 42 secondary); conditions are not pooled so no overall significance is claimed; CIs for some practically-equivalent verdicts have lower bounds close to the margin.
6. Missing data. A1 in S3 at M = 3 excluded (2,640 of 2,700 runs); the excluded arm-condition cannot be compared.
7. Provenance. The held-out evaluation used the tracked code snapshot identified by commit 7a81352 in the run metadata. The v0.7 tag points to an earlier protocol commit (f874ad7), so the evaluated code is identified by the commit rather than by the tag.

9 Future Work

(a) Component ablation of A5 (E-only, +C, +V, fixed vs adaptive Θ, feedback on/off) to test whether any single signal helps; (b) hybrid policies that front-load and then adapt; (c) per-seed matched mutation counts; (d) additional scenario families, real failure traces and larger scale; (e) genuinely distributed controllers and delayed information; (f) implementing and testing load, traffic and trust signals, and a meaningful "resilience cost" for M; (g) power analysis for smaller effects.

10 Conclusion

Across six evaluated simulated conditions, full TCCE-R did not establish an advantage over the reactive baseline, while front-loaded adaptation achieved lower consensus error in every condition. This bounded result does not refute the individual resilience signals; it identifies the need for component ablations and broader validation.

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Appendices

Appendix A. Full secondary comparisons

Condition	Comparison (E_X − E_Y)	n	ΔE	95% bootstrap CI	Holm p	Reading
S1, M=3	A2 − A1	60	-0.0006	[-0.0072, +0.0052]	1.000	practically equivalent
S1, M=6	A2 − A1	60	-0.0031	[-0.0110, +0.0047]	1.000	practically equivalent
S2, M=3	A2 − A1	60	+0.0040	[-0.0033, +0.0113]	1.000	practically equivalent
S2, M=6	A2 − A1	60	-0.0050	[-0.0133, +0.0030]	1.000	practically equivalent
S3, M=6	A2 − A1	60	-0.0041	[-0.0076, -0.0008]	0.690	practically equivalent
S1, M=3	A3 − A1	60	-0.0081	[-0.0150, -0.0015]	0.732	practically equivalent
S1, M=6	A3 − A1	60	-0.0141	[-0.0214, -0.0072]	0.058	inconclusive
S2, M=3	A3 − A1	60	-0.0058	[-0.0134, +0.0016]	1.000	practically equivalent

S2, M=6	A3 − A1	60	-0.0044	[-0.0110, +0.0020]	1.000	practically equivalent
S3, M=6	A3 − A1	60	-0.0038	[-0.0066, -0.0012]	0.128	practically equivalent
S1, M=3	A3 − A2	60	-0.0075	[-0.0133, -0.0021]	0.456	practically equivalent
S1, M=6	A3 − A2	60	-0.0110	[-0.0193, -0.0030]	0.060	practically equivalent
S2, M=3	A3 − A2	60	-0.0098	[-0.0178, -0.0023]	0.315	practically equivalent
S2, M=6	A3 − A2	60	+0.0006	[-0.0062, +0.0076]	1.000	practically equivalent
S3, M=3	A3 − A2	60	-0.0038	[-0.0064, -0.0014]	0.128	practically equivalent
S3, M=6	A3 − A2	60	+0.0002	[-0.0034, +0.0039]	1.000	practically equivalent
S1, M=3	A4 − A3	60	+0.0021	[-0.0029, +0.0075]	1.000	practically equivalent
S1, M=6	A4 − A3	60	+0.0105	[+0.0038, +0.0170]	0.018	A3 lower E, < margin
S2, M=3	A4 − A3	60	+0.0144	[+0.0075, +0.0216]	0.003	A3 lower E, < margin
S2, M=6	A4 − A3	60	+0.0090	[+0.0043, +0.0139]	0.057	practically equivalent
S3, M=3	A4 − A3	60	+0.0008	[-0.0013, +0.0029]	1.000	practically equivalent
S3, M=6	A4 − A3	60	+0.0038	[+0.0006, +0.0069]	0.307	practically equivalent
S1, M=3	A5 − A4	60	+0.0007	[-0.0039, +0.0050]	1.000	practically equivalent
S1, M=6	A5 − A4	60	-0.0009	[-0.0052, +0.0034]	1.000	practically equivalent
S2, M=3	A5 − A4	60	+0.0019	[-0.0033, +0.0074]	1.000	practically equivalent
S2, M=6	A5 − A4	60	+0.0034	[-0.0008, +0.0086]	1.000	practically equivalent
S3, M=3	A5 − A4	60	+0.0020	[-0.0000, +0.0039]	0.307	practically equivalent
S3, M=6	A5 − A4	60	+0.0002	[-0.0017, +0.0023]	1.000	practically equivalent
S1, M=3	A6 − A1	60	-0.0241	[-0.0346, -0.0136]	0.002	A6 lower E, ≥ margin
S1, M=6	A6 − A1	60	-0.0357	[-0.0484, -0.0229]	1.42×10^{-4}	A6 lower E, ≥ margin
S2, M=3	A6 − A1	60	+0.0019	[-0.0109, +0.0149]	1.000	practically equivalent
S2, M=6	A6 − A1	60	+0.0245	[+0.0095, +0.0401]	0.091	inconclusive
S3, M=6	A6 − A1	60	-0.0142	[-0.0204, -0.0079]	0.003	A6 lower E, < margin

S1, M=3	A7 − A1	60	-0.0501	[-0.0647, -0.0348]	1.85×10^{-5}	A7 lower E, ≥ margin
S1, M=6	A7 − A1	60	-0.0802	[-0.0949, -0.0651]	5.73×10^{-9}	A7 lower E, ≥ margin
S2, M=3	A7 − A1	60	-0.0356	[-0.0515, -0.0191]	0.003	A7 lower E, ≥ margin
S2, M=6	A7 − A1	60	-0.0302	[-0.0460, -0.0129]	0.034	A7 lower E, ≥ margin
S3, M=6	A7 − A1	60	-0.0319	[-0.0382, -0.0253]	4.04×10^{-8}	A7 lower E, ≥ margin

Appendix B. Frozen calibration report

Condition	Arm	Threshold	Mean count (calibration seeds)	Feasible (±15% of M)
S1, M=3	A1	0.753	3.40	yes
S1, M=3	A2	0.701	3.00	yes
S1, M=3	A3	1.712	2.80	yes
S1, M=3	A4	2.714	3.00	yes
S1, M=3	A5	3.165	2.90	yes
S1, M=6	A1	0.701	6.05	yes
S1, M=6	A2	0.631	5.85	yes
S1, M=6	A3	0.681	6.25	yes
S1, M=6	A4	1.259	6.15	yes
S1, M=6	A5	1.259	6.25	yes
S2, M=3	A1	0.649	3.10	yes
S2, M=3	A2	0.562	2.90	yes
S2, M=3	A3	0.501	2.90	yes
S2, M=3	A4	0.794	3.00	yes
S2, M=3	A5	0.926	2.90	yes
S2, M=6	A1	0.527	6.05	yes
S2, M=6	A2	0.439	6.00	yes
S2, M=6	A3	0.108	6.20	yes
S2, M=6	A4	0.068	5.70	yes
S2, M=6	A5	0.079	6.15	yes
S3, M=3	A1	0.753	2.50	NO
S3, M=3	A2	0.701	2.75	yes
S3, M=3	A3	1.468	3.00	yes
S3, M=3	A4	3.165	3.00	yes
S3, M=3	A5	3.691	3.00	yes
S3, M=6	A1	0.701	6.55	yes
S3, M=6	A2	0.649	6.45	yes
S3, M=6	A3	0.681	6.25	yes
S3, M=6	A4	1.712	5.70	yes
S3, M=6	A5	1.712	6.45	yes

Appendix C. Mechanism check (exploratory)

Condition	Arm	Mutations (all runs)	Made while disconnected	Reconnecting moves	Mean λ_2-gain estimate (connected)	Mean epoch of first mutation
S1, M=3	A1	148	99.3%	94.6%	0.161	24.3
S1, M=3	A3	143	81.8%	74.1%	0.153	24.4
S1, M=3	A5	150	93.3%	84.0%	0.150	24.2
S1, M=3	A6	135	31.9%	27.4%	0.147	24.0
S1, M=3	A7	135	15.6%	8.1%	0.148	0.0
S1, M=6	A1	325	97.2%	84.0%	0.161	13.8
S1, M=6	A3	318	85.2%	71.7%	0.155	13.3
S1, M=6	A5	325	94.2%	78.5%	0.163	13.9
S1, M=6	A6	288	31.2%	27.8%	0.118	14.3
S1, M=6	A7	288	23.6%	14.6%	0.119	0.0
S2, M=3	A1	151	72.2%	64.2%	0.157	17.2
S2, M=3	A3	144	61.1%	53.5%	0.158	16.5
S2, M=3	A5	140	82.1%	75.7%	0.173	21.3
S2, M=3	A6	132	32.6%	30.3%	0.155	24.1
S2, M=3	A7	132	6.8%	6.8%	0.153	0.0
S2, M=6	A1	313	45.4%	38.3%	0.148	6.2
S2, M=6	A3	317	41.3%	34.7%	0.149	5.4
S2, M=6	A5	312	46.8%	39.4%	0.155	8.7
S2, M=6	A6	292	20.2%	17.8%	0.128	13.2
S2, M=6	A7	292	8.9%	7.5%	0.121	0.0
S3, M=3	A3	160	85.6%	80.6%	0.078	21.9
S3, M=3	A5	158	97.5%	88.6%	0.115	26.1
S3, M=3	A6	142	34.5%	31.7%	0.079	23.7
S3, M=3	A7	142	23.2%	19.7%	0.086	0.0
S3, M=6	A1	330	97.0%	87.9%	0.077	15.0
S3, M=6	A3	325	89.2%	80.0%	0.075	13.3
S3, M=6	A5	334	97.0%	83.8%	0.069	15.8
S3, M=6	A6	307	44.6%	39.1%	0.067	13.6
S3, M=6	A7	307	29.6%	23.1%	0.073	0.0

Appendix D. Placebo / regression-to-mean check (v0.6 data; exploratory)

v0.6 condition	Feedback updates	Legacy rule, A5's own updates: success	Legacy rule on Static run (no mutation): 'success'
d2_M3	152	80.9%	67.8%
d2_M6	313	79.6%	72.8%
d8_M3	141	77.3%	67.4%
d8_M6	295	79.0%	72.5%

Appendix E. Reproducibility summary

Evaluation: code snapshot identified by commit 7a81352 (git description v0.7-2-g7a81352), run on a clean tracked tree with a frozen 47-file manifest; Python 3.11.9, NumPy 2.3.4, SciPy 1.17.1, and pandas 2.3.3; 2,640 runs in 272.3 seconds. Seeds: calibration 200–219, development 300–319, and held-out evaluation 7000–7059. The accompanying analysis scripts regenerate the tables and figures and check numerical claims.